

Question

Isotope labeling is a classical method used to study metabolic processes in living organisms. We consider culturing mammalian cells under oxygen-sufficient conditions using glucose labeled with different ^{13}C , and adding different exogenous molecules to regulate cellular metabolism. Below is a description of the corresponding experimental methods and expected results. Please select the correct one.

- A.** 1. Cultivate cells with glucose labeled at the C4 position with ^{13}C , without adding any other exogenous small molecules, and the first carbon in pyruvate to be isotopically labeled will be the keto carbonyl carbon.
- B.** After pretreating cells with iodoacetate, the culture medium was removed. When cells were cultured with glucose labeled with ^{13}C at the C6 position, the final isotopic labeling remained on the aldehyde carbon of glyceraldehyde-3-phosphate or the phosphate carbon of dihydroxyacetone phosphate.
- C.** After pretreating the cells with sodium arsenate, the culture medium was removed. Cells were then cultured with glucose labeled with ^{13}C at the C3 position, without adding any other exogenous small molecules, resulting in pyruvate not being isotopically labeled.
- D.** When cells are cultured with glucose labeled at the C6 position using ^{13}C and a certain concentration of sodium fluoride is added to the culture medium, the final isotopic labeling is concentrated on the methylene carbon of phosphoenolpyruvate.
- E.** After culturing cells with glucose labeled at the C4 position with ^{13}C , the cells were collected within a short period, and saturated fatty acids were isolated from the cells. The ^{13}C isotopic label could be detected in these fatty acids.
- F.** After culturing cells with glucose labeled at the C1 position with ^{13}C , isolating the intracellular ketone bodies and detecting the isotopic labeling, the isotopic label should appear in the methyl and methylene groups of 3-hydroxybutyrate and acetoacetate, as well as in the carbonyl carbon of acetone.

- G.** Cells were cultured with glucose labeled at the C6 position with ^{13}C , and a high concentration of fluoroacetate was added to the culture medium. This treatment did not affect the detection of isotopic labeling in oxaloacetate.
- H.** When cells are cultured with glucose labeled with ^{13}C at the C3 position, the isotopic label can be detected in the carbon dioxide produced by cellular metabolism after at least one round of the citric acid cycle.
- I.** When cells are cultured with glucose labeled with ^{13}C at the C6 position, a minimum of 3 rounds of the citric acid cycle are required to generate isotopically labeled carbon dioxide in the citric acid cycle.
- J.** None of the above statements are correct

Answer

Correct Answer: I

Detailed Explanation

First, the glycolysis process is examined. In glycolysis, the C1-C6 of glucose remain unchanged until the formation of fructose-1,6-bisphosphate. Subsequently, fructose-1,6-bisphosphate undergoes cleavage: carbons 1, 2, and 3 enter dihydroxyacetone phosphate (DHAP), with carbon 1 bonded to a phosphate group, carbon 2 as a carbonyl group, and carbon 3 bonded to a hydroxyl group; carbons 4, 5, and 6 enter glyceraldehyde-3-phosphate (G3P), with carbon 4 as an aldehyde group, carbon 5 bonded to a hydroxyl group, and carbon 6 bonded to a phosphate group. Under the action of triosephosphate isomerase, dihydroxyacetone phosphate isomerizes to glyceraldehyde-3-phosphate. This process does not involve the migration of the phosphate group, so carbon 1 retains the phosphate group, carbon 2 is bonded to a hydroxyl group, and carbon 3 remains an aldehyde group. Therefore, in glucose labeling, after the formation of glyceraldehyde-3-phosphate, the pairs 1/6, 2/5, and 3/4 are equivalent within their respective groups.

CHECKPOINT

1 PTS

In isotopically labeled glucose, the labeling can be divided into three cases: 1/6, 2/5, and 3/4, where the two labels within each group are equivalent.

First, consider the 3/4 group, corresponding to options A, C, E, and H. The aldehyde group of glyceraldehyde-3-phosphate corresponds to the carboxyl group in pyruvate, carbon 5 corresponds to the keto carbonyl group, and carbon 6 corresponds to the methyl group. Thus, the isotope in pyruvate appears in the carboxyl group. Option A is incorrect.

CHECKPOINT

1 PTS

Labeling glucose at C4 results in the isotope appearing in the carboxyl group of pyruvate; A is incorrect.

Sodium arsenate is an inhibitor of glyceraldehyde-3-phosphate oxidation, capable of uncoupling oxidation and phosphorylation. However, it is a reversible inhibitor, so glycolysis resumes normally after removal from the culture medium. Therefore, option C is incorrect.

CHECKPOINT

1 PTS

Sodium arsenate inhibition is reversible, and glycolysis can resume normally after removal; C is incorrect.

CHECKPOINT

1 PTS

The isotope first appears in the carboxyl group of pyruvate.

When pyruvate is converted to acetyl-CoA, the carboxyl group is released as carbon dioxide. Thus, the isotope does not appear in downstream metabolic processes, making options E and H incorrect.

CHECKPOINT

1 PTS

The isotopic label on the carboxyl group of pyruvate does not enter acetyl-CoA and thus will not appear in synthesized saturated fatty acids; E is incorrect.

CHECKPOINT

1 PTS

The isotope does not need to enter the citric acid cycle to produce isotopically labeled carbon dioxide via acetyl-CoA synthesis; H is incorrect.

Next, examine the 1/6 group, corresponding to descriptions 2, 4, 6, 7, 9, and 10. Iodoacetate is an irreversible inhibitor of GAPDH, and its inhibition cannot be reversed by removing the culture medium. Glycolysis ultimately halts at glyceraldehyde-3-phosphate (or dihydroxyacetone phosphate), with the isotopic label on the carbon bonded to the phosphate group. Option B is incorrect.

CHECKPOINT

1 PTS

Iodoacetate is an irreversible inhibitor of GAPDH.

CHECKPOINT

1 PTS

The isotopic label in both compounds is located on the carbon bonded to the phosphate group; B is incorrect.

Sodium fluoride inhibits pyruvate kinase and enolase, causing glycolysis to halt at 2-phosphoglycerate rather than accumulating phosphoenolpyruvate. The isotopic label is on the methylene carbon, making option D incorrect.

CHECKPOINT

1 PTS

Sodium fluoride inhibits pyruvate kinase and enolase.

CHECKPOINT

1 PTS

The isotopic label primarily remains at 2-phosphoglycerate; D is incorrect.

In acetyl-CoA, the methyl carbon carries the isotopic label, which subsequently enters the ketone body synthesis pathway. Three molecules of acetyl-CoA synthesize HMG-CoA, where the isotopic label appears on the α -carbon of the thioester, the β -methyl carbon, and the α -carbon of the carboxyl group. HMG-CoA is then cleaved to produce acetoacetate, with the isotopic label appearing on the methyl and methylene groups. Acetoacetate decarboxylates to form acetone, with the label on both methyl groups; or is reduced to 3-hydroxybutyrate, where the label remains on the methyl and methylene groups. Option F is incorrect.

CHECKPOINT

1 PTS

In ketone bodies synthesized from methyl-labeled acetyl-CoA, the isotope is located on methyl or methylene groups; F is incorrect.

Fluoroacetate is converted to fluoroacetyl-CoA, which reacts with oxaloacetate to form fluorocitrate, inhibiting citrate synthase and blocking the citric acid cycle. Since isotopically labeled glucose requires glycolysis to produce acetyl-CoA, and large amounts of fluoroacetate can block the citric acid cycle early on, labeled acetyl-CoA hardly enters the cycle. Thus, oxaloacetate shows almost no isotopic labeling. Option G is incorrect.

CHECKPOINT

1 PTS

Fluoroacetate-derived fluorocitrate inhibits citrate synthase.

CHECKPOINT

1 PTS

The citric acid cycle is blocked, so oxaloacetate will not show isotopic labeling.

Finally, examine isotopic transfer in the citric acid cycle. When the methyl group of acetyl-CoA carries an isotopic label, it appears on one of the methylene groups of succinate in the cycle. Note that succinate is a symmetric molecule; after hydration to malate and oxidation to oxaloacetate, the isotopic label may appear on either the carbonyl or methylene group of oxaloacetate. If the methylene group is labeled, in the next cycle, the label will appear on the carboxyl group of oxaloacetate. If the carbonyl group is labeled, in the next cycle, the label will appear on the hydroxyl carbon of citrate, return to the methylene group of succinate, and then reappear on the carbonyl or methylene group of oxaloacetate. Considering only the carboxyl group of oxaloacetate, one carboxyl group is released as CO_2 during the conversion of α -ketoglutarate to succinyl-CoA. Thus, option I is correct.

CHECKPOINT

1 PTS

At least three rounds of the citric acid cycle are required: acetyl-CoA $\rightarrow$ methylene of oxaloacetate $\rightarrow$ carboxyl of oxaloacetate $\rightarrow \text{CO}_2$, to transfer the isotope to CO_2 ; option I is correct.